

AI & IOT INTEGRATED WIND TREE FOR OPTIMIZED MICRO WIND ENERGY

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IN**

Computer Science & Engineering

Submitted by

M.THANMAI	22491A05G6
G.SRI SOWMYA	22491A05E0
G.AKSHA	22491A0583
K.MOULYA SRI	22491A05E9
K.ZUBAIR	22491A05E8
K.NAGA JYOTHI	22491A05F8

Under the esteemed guidance of

Dr.Vidyanadha babu

Associate Professor



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

QIS COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

*An ISO 9001:2015 Certified institution, approved by AICTE & Reaccredited by NBA, NAAC 'A+' Grade
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VENGAMUKKAPALEM, ONGOLE – 523 272, A.P

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VENGAMUKKAPALEM, ONGOLE:-523272, A.P

December 2025



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the *technical report* entitled **“AI & IOT INTEGRATED WIND TREE FOR OPTIMIZED MICRO WIND ENERGY”** is a bonafide work of the following final B Tech students in the partial fulfillment of the requirement for the award of the degree of bachelor of technology in *COMPUTER SCIENCE & ENGINEERING* for the academic year 2025-2026.

M.THANMAI	22491A05G6
G.SRI SOWMYA	22491A05E0
G.AKSHA	22491A0583
K.MOULYA SRI	22491A05E9
K.ZUBAIR	22491A05E8
K.NAGA JYOTHI	22491A05F8

Signature of the guide

Dr.Vidyanadha babu, M.Tech, Ph.D

Associate Professor

Signature of Head of Department

Dr. Buchi babu, M.Tech, Ph.D

HOD, Professor in CSE

Signature of External Examiner

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Submitted by

M.Thanmai	(22491A05G6)
G.Sri Sowmya	(22491A05E0)
G.Aksha	(22491A0583)
K.Moulya Sri	(22491A05E9)
K.Naga Jyothi	(22491A05F8)
K.Zubair	(22491A05E8)

ABSTRACT

The project “**AI & IoT Integrated Wind Tree for Optimized Micro Wind Energy**” presents an innovative approach to generating renewable electricity by combining bio-inspired design with modern digital technologies. The objective is to harness low-speed wind through **aero leaves**, which act as miniature vertical axis wind turbines (VAWTs), and optimize power generation using IoT and AI integration.

The methodology involves the deployment of **aero leaves with micro DC generators** to capture wind energy and convert it into electrical power. The generated electricity is rectified, regulated by charge controllers, and stored in batteries for later use. An **IoT-enabled microcontroller (ESP32/NodeMCU)** collects data from environmental and electrical sensors such as anemometers, voltage/current sensors, and temperature/humidity modules. This data is transmitted to cloud platforms (Thingspeak, Blynk, or custom dashboards) for real-time monitoring. An **AI-based predictive model** is trained on historical datasets of wind flow and power output to forecast future energy availability, optimize battery usage, and suggest smart utilization strategies.

Key findings demonstrate that the system is effective in low-wind environments, making it suitable for **urban landscapes, parks, and rural communities**. The integration of IoT enables continuous monitoring and remote accessibility, while AI improves prediction accuracy and operational efficiency, ensuring optimized energy harvesting at micro levels.

In conclusion, the proposed wind tree system delivers a **sustainable, scalable, and intelligent solution** for micro wind energy generation. By merging AI, IoT, and renewable technology, this project contributes to **green energy innovation, smart city infrastructure, and decentralized power generation**.

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LIST OF SYMBOLS AND ABBREVIATIONS

- **AI** – Artificial Intelligence
- **IoT** – Internet of Things
- **VAWT** – Vertical Axis Wind Turbine
- **DC** – Direct Current
- **AC** – Alternating Current
- **MQTT** – Message Queuing Telemetry Transport
- **HTTP** – HyperText Transfer Protocol
- **DHT** – Digital Humidity & Temperature Sensor
- **ESP32** – Espressif Systems Microcontroller
- **UI** – User Interface
- **Wi-Fi** – Wireless Fidelity
- **ML** – Machine Learning

1. Introduction

1.1 INTRODUCTION

The project “**AI & IoT Integrated Wind Tree for Optimized Micro Wind Energy**” is an innovative renewable energy system designed to harness wind power efficiently, even at low wind speeds. The wind tree is equipped with multiple **aero leaves**, which act as miniature vertical axis wind turbines that rotate with the wind and generate electricity through micro DC generators. The generated power is rectified, regulated, and stored in batteries for later use. To make the system intelligent, **IoT sensors** such as anemometers, voltage and current sensors, and environmental monitors are integrated to collect real-time data on wind conditions and energy output. This data is transmitted via Wi-Fi-enabled microcontrollers to cloud platforms for monitoring and control. Additionally, an **AI-based prediction model** is used to analyze historical wind and energy data, forecast energy availability, and optimize power usage. The stored energy can be utilized for small-scale applications such as LED lighting, mobile charging, and microgrid support, making the system suitable for urban environments, smart cities, and rural areas. By combining **sustainability, smart technology, and bio-inspired design**, this project offers a compact, efficient, and eco-friendly solution for decentralized power generation.

1.2 OBJECTIVE OF THE PROJECT

The main objective of the project “**AI & IoT Integrated Wind Tree for Optimized Micro Wind Energy**” is to design and develop a smart, sustainable energy system that can harness low-speed wind efficiently. The specific objectives are:

1. To design and implement a **wind tree with aero leaves** that can capture low-speed wind and generate electricity through micro DC generators.
2. To integrate **IoT sensors and microcontrollers** for real-time monitoring of wind speed, environmental conditions, and power output.
3. To develop an **AI-based predictive model** to forecast wind energy availability and optimize energy storage and utilization.
4. To ensure efficient **storage and regulation** of generated electricity using rectifiers, charge controllers, and batteries.
5. To demonstrate the application of stored power in **small-scale utilities** such as lighting, charging, and microgrids.
6. To contribute towards **green technology, smart city infrastructure, and decentralized renewable energy solutions**.

1.3 MOTIVATION OF THE THESIS

The increasing global demand for energy, coupled with the harmful impacts of fossil fuels on the environment, has created an urgent need for sustainable and clean energy solutions. While large-scale wind and solar projects are already contributing significantly, they are often unsuitable for **urban and semi-urban environments** due to space requirements, noise, and high installation costs. Moreover, many rural and remote areas still lack reliable access to electricity. These challenges motivate the development of a **compact, efficient, and aesthetically pleasing micro wind energy system** that can work even under low wind conditions.

The concept of a **wind tree with aero leaves** offers a promising solution by combining bio-inspired design with modern technologies. By integrating **IoT for real-time monitoring** and **AI for predictive analysis and optimization**, the system not only generates renewable electricity but also ensures smart utilization and efficiency. The motivation behind this thesis is to bridge the gap between traditional renewable energy systems and **next-generation intelligent energy solutions**, providing communities with an eco-friendly, decentralized, and sustainable power source that supports both **green energy adoption and smart city development**.

1.4 ORGANISATION OF THESIS

This thesis is organized into five chapters, each addressing a specific aspect of the project:

- **Chapter 1: Introduction**
Provides the background, motivation, problem statement, objectives, and scope of the project. It sets the context and explains the importance of developing an AI & IoT integrated wind tree for optimized micro wind energy.
- **Chapter 2: Literature Review**
Reviews previous research and related works in the areas of renewable energy, micro wind turbines, IoT-based energy systems, and AI applications in energy prediction and optimization. This chapter highlights the research gap that the present work aims to address.
- **Chapter 3: Methodology**
Describes the system design and methodology adopted in the project. It includes details of hardware components such as aero leaves, generators, sensors, and batteries, as well as software tools like Arduino IDE, IoT platforms, and AI models.

- **Chapter 4: Results and Discussion**

Presents the implementation results, data analysis, and performance evaluation of the proposed system. This chapter discusses how IoT integration and AI prediction improve efficiency and compares outcomes with conventional approaches.

- **Chapter 5: Conclusion and Future Work**

Summarizes the overall findings, conclusions drawn from the study, and the contributions of the project. It also outlines the potential improvements and future directions for enhancing the system's performance and scalability.

2. Literature Review / Background

2.1 BASIC CONCEPT

The foundation of this project lies in the integration of **renewable energy systems, Internet of Things (IoT), and Artificial Intelligence (AI)** to develop a smart micro wind energy solution. A brief overview of the basic concepts is given below:

1. Micro Wind Energy Systems

Wind energy is one of the most reliable and mature renewable energy sources. While conventional large-scale horizontal axis wind turbines (HAWTs) are widely used, they are often unsuitable for low-wind urban environments due to noise, size, and high installation requirements. **Vertical Axis Wind Turbines (VAWTs)**, particularly the Savonius type, offer an alternative as they can operate efficiently at lower wind speeds and in turbulent conditions. The concept of **aero leaves** builds on this principle by miniaturizing VAWTs into leaf-like structures mounted on a tree framework.

2. Electricity Generation and Storage

The mechanical rotation of the aero leaves is converted into electrical energy using **micro DC generators**. Since the generated output is unstable, **bridge rectifiers** and **charge controllers** are used for conversion and regulation before storing the energy in batteries. This ensures a consistent and safe supply of renewable electricity for small-scale applications.

3. Internet of Things (IoT)

IoT plays a vital role in monitoring and managing renewable energy systems. By integrating sensors such as **anemometers, voltage/current sensors, and environmental monitors**, real-time data can be collected and transmitted via microcontrollers like **ESP8266/ESP32**. This data is visualized using platforms such as **Thingspeak, Blynk, or Firebase**, enabling remote monitoring, fault detection, and performance optimization.

4. Artificial Intelligence (AI) in Energy Systems

AI enhances the efficiency of renewable energy by providing **prediction and optimization capabilities**. By applying machine learning algorithms such as **linear regression, random forest, or LSTM networks**, wind patterns can be forecasted, and energy utilization can be optimized. This reduces energy wastage, improves reliability, and enables predictive maintenance of the wind tree system.

5. Integration of AI and IoT (AIoT)

The combination of AI and IoT, often referred to as **AIoT**, makes the system intelligent and adaptive. While IoT ensures connectivity and real-time data

acquisition, AI provides decision-making and forecasting abilities. Together, they transform the wind tree into a **smart renewable energy solution** capable of self-optimization and remote accessibility.

2.2 RELATED WORK

A number of studies have explored renewable energy systems, particularly wind power, and their integration with digital technologies such as IoT and AI. This section summarizes and analyzes the existing research relevant to the project.

Micro Wind Energy Systems

Researchers have investigated small-scale wind turbines, particularly **Vertical Axis Wind Turbines (VAWTs)**, for decentralized energy generation. Studies show that Savonius-type VAWTs are more effective in **low-speed, turbulent urban wind conditions** compared to horizontal axis turbines. However, their efficiency remains limited, and scaling them into modular units, such as **aero leaves on a wind tree structure**, has not been widely explored in practice.

Electricity Storage and Management

Several works emphasize the importance of **energy storage** in renewable systems. Techniques involving batteries, charge controllers, and rectifiers are well established. Prior studies highlight challenges in **overcharging, unstable outputs, and poor energy utilization efficiency**, which motivates the need for smarter regulation and predictive energy management.

IoT-Based Energy Monitoring

IoT integration in renewable energy systems has gained popularity in recent years. Projects using **ESP8266/ESP32 microcontrollers** and platforms like **Thingspeak, Blynk, and Node-RED** demonstrate real-time monitoring of voltage, current, and environmental data. However, most of these solutions focus only on **data acquisition and visualization**, lacking intelligent decision-making capabilities.

Artificial Intelligence in Renewable Energy

AI techniques such as **machine learning (ML) and deep learning models** have been applied for wind forecasting, energy prediction, and fault detection. Research shows that models like **Linear Regression, Random Forest, and LSTM networks** can provide reliable predictions of wind flow and energy availability. Despite this, many implementations are limited to simulations or large-scale wind farms, with few focusing on **micro wind energy systems**.

AI + IoT (AIoT) in Smart Energy Systems

Emerging works combine IoT data collection with AI-based analysis to create **intelligent renewable systems**. This AIoT approach improves efficiency, enables predictive maintenance, and optimizes energy usage. However, a clear gap exists in applying AIoT to **bio-inspired micro wind systems** such as wind trees, where scalability, cost-effectiveness, and low-wind performance remain underexplored.

Research Gap and Contribution

While existing literature demonstrates progress in micro wind turbines, IoT-based monitoring, and AI-driven prediction, there is a lack of integrated systems that combine all three for **low-wind urban and rural environments**. This project aims to fill this gap by developing an **AI & IoT integrated wind tree** with aero leaves that not only generates power but also **monitors, predicts, and optimizes** energy usage in real time.

2.3 RESEARCH GAP

Although several studies have been conducted in the fields of **renewable energy, IoT-based monitoring, and AI-driven prediction systems**, there are still important gaps that this project aims to address:

- 1. Limited focus on micro wind systems for low-wind conditions**
Most existing wind energy projects concentrate on large-scale horizontal axis turbines or high-speed wind environments. Very few solutions are designed specifically for **urban and semi-urban areas** where wind speeds are relatively low and irregular.
- 2. Lack of integration between energy generation, IoT, and AI**
Previous research often treats these areas separately—either focusing on micro wind turbine design, IoT-based monitoring, or AI prediction models. Comprehensive systems that **combine AI and IoT with micro wind energy harvesting** are still limited.
- 3. Aesthetic and space-related challenges in urban renewable systems**
Conventional turbines are often unsuitable for cities due to their **size, noise, and visual impact**. While bio-inspired concepts like wind trees exist, there is limited research on practical, cost-effective, and optimized designs with **functional aero leaves** for electricity generation.
- 4. Inefficient energy management in existing micro wind projects**
Many small-scale renewable setups generate power but lack **predictive analytics and smart load management**, leading to underutilization of stored energy and reduced system efficiency.

5. **Limited deployment of AI in decentralized renewable systems**

AI-based prediction models have mostly been applied to large wind farms, while **micro-scale AIoT-enabled renewable systems** are underexplored, especially for localized applications such as lighting, charging, and microgrids.

IDENTIFIED GAP

There is a clear need for a **compact, intelligent, and scalable micro wind energy system** that can operate effectively in **low-wind urban and rural settings**, integrate **IoT for real-time monitoring**, and utilize **AI for prediction and optimization**.

3. PROPOSED WORK AND ANALYSIS

3.1 PROPOSED SYSTEM

The proposed system is a **Smart Wind Energy Harvesting and Monitoring System** that integrates renewable energy capture with IoT-based monitoring, cloud connectivity, and AI-driven predictions for optimized energy usage.

1. Energy Capture and Conversion

- Vertical Axis Wind Turbine (VAWT) blades are used to harness wind energy efficiently, even at low wind speeds.
- Micro DC generators convert the mechanical energy into electrical energy.
- A bridge rectifier converts AC to DC, and a charge controller regulates voltage and current before storing energy in a battery bank.

2. IoT Monitoring and Control

- Sensors such as anemometer (wind speed), voltage/current sensors, and DHT11/DHT22 (temperature and humidity) continuously monitor environmental and system parameters.
- An ESP32/NodeMCU microcontroller processes the sensor data and controls the charging process.

3. Data Transmission and Cloud Integration

- The system uses Wi-Fi connectivity to transmit real-time data to cloud platforms such as **ThingSpeak** or **Firebase**.
- Communication protocols like **MQTT/HTTP** ensure reliable and secure data transfer.

4. AI Prediction and Optimization

- Optimal energy storage and utilization strategies are suggested to improve efficiency and reduce wastage.

5. User Interface

- A web/mobile dashboard is developed to provide real-time visualization of power generation, battery status, environmental data, and AI-based predictions.

- The dashboard helps users make informed decisions on energy consumption and storage management.

Advantages of Proposed System

1. Efficient energy harvesting using VAWT suitable for variable wind speeds.
2. Real-time monitoring and control via IoT.
3. Cloud-based data storage and access from anywhere.
4. AI-driven predictions enhance reliability and planning.
5. User-friendly dashboard for energy tracking and decision-making.

Module	Key Components	Function
Energy Capture	VAWT Blades, Micro DC Generator	Capture wind energy and convert it to AC power.
Conversion & Storage	Rectifier, Charge Controller, Battery Bank	Convert AC to DC, regulate, and store power.
IoT Monitoring	Anemometer, V/I Sensor, DHT11/22, ESP32	Monitor wind, environment, voltage/current; process data.
Data Transmission	Wi-Fi (ESP32), Cloud (ThingSpeak/Firebase)	Send data to cloud for storage and access.
AI Prediction	ML Models (LR, RF, LSTM)	Predict wind patterns and optimize energy usage.
User Interface	Web/Mobile Dashboard	Display real-time data and AI predictions.

Table1:Required components

3.2 Feasibility study

The proposed AI & IoT Integrated Wind Tree is a highly feasible solution for optimized micro wind energy generation. From a technical feasibility point of view, the system makes use of well-established and easily available components such as vertical-axis wind turbines designed as tree-like aero leaves, micro DC generators, bridge rectifiers, charge controllers, ESP32/NodeMCU microcontrollers, and sensors like anemometers and DHT11/DHT22. The integration of IoT through Wi-Fi and cloud platforms such as ThingSpeak or Firebase ensures reliable data transfer and storage, while AI algorithms including Linear Regression, Random Forest, and LSTM provide accurate predictions of wind patterns and energy output, proving the system to be technically sound.

In terms of **economic feasibility**, the system requires an initial investment for turbines, sensors, batteries, and communication modules, but the long-term benefits outweigh the cost.

The running expenses are minimal, limited to periodic maintenance and battery replacement, while savings from reduced electricity bills and the possibility of supplying surplus energy to the grid ensure financial viability with a short payback period.

The **operational feasibility** of the wind tree is strong, as it is designed for user-friendliness. Real-time monitoring and control are achieved through a mobile or web dashboard, making it simple for users to track performance. Automation through IoT and AI reduces human intervention, while predictive insights optimize storage and energy usage, enhancing reliability.

The system is also **environmentally feasible**, as it utilizes renewable wind energy, produces zero emissions, and contributes to sustainable development goals. Its compact “tree-like” structure makes it aesthetically suitable for both urban and rural areas, without requiring large land spaces like traditional wind turbines.

Lastly, from a **social feasibility** perspective, the wind tree has significant benefits. It promotes clean energy adoption, increases awareness of smart renewable technologies, and provides reliable electricity in areas with limited grid access. This makes it both socially acceptable and beneficial to communities.

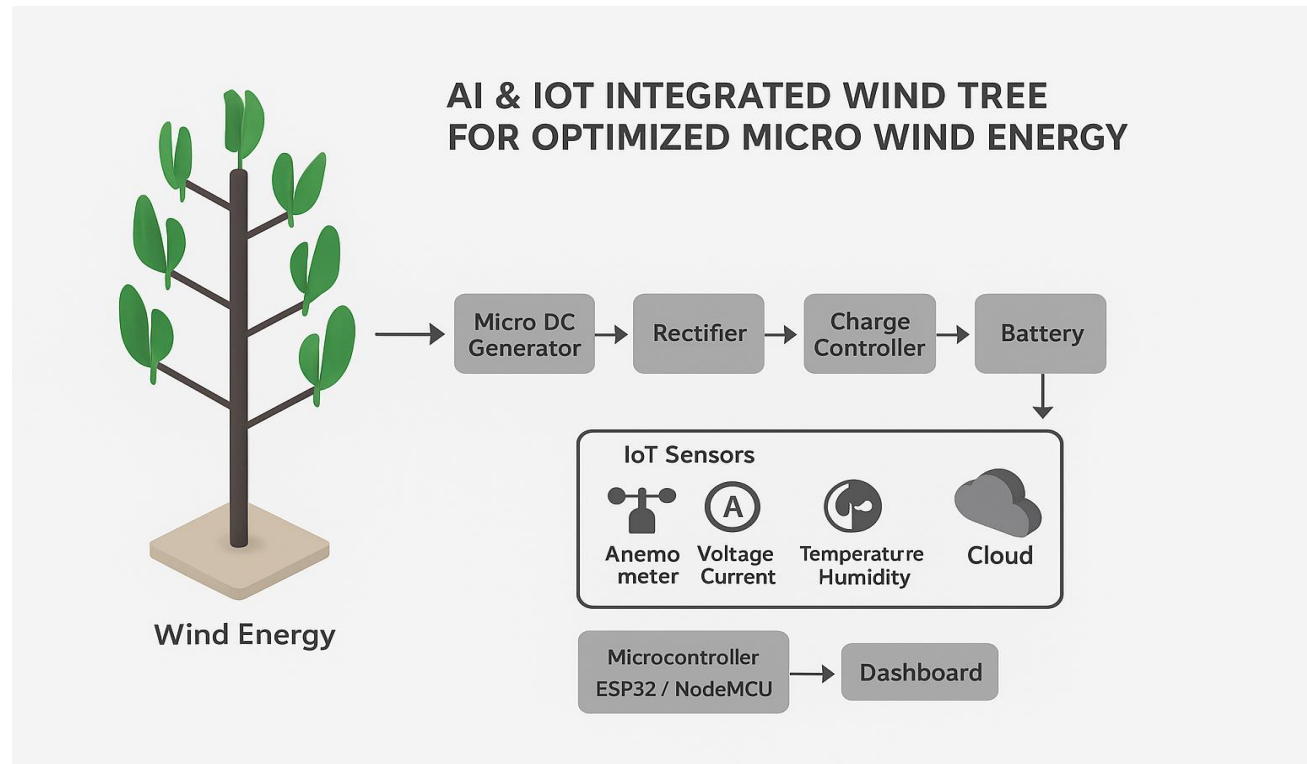
In conclusion, the proposed AI & IoT Integrated Wind Tree is technically sound, economically viable, operationally practical, environmentally sustainable, and socially impactful, making it a highly feasible and innovative solution for optimized micro wind energy generation.

4 SYSTEM DESIGN

4.1 SYSTEM DESIGN

The proposed system “**AI & IoT Integrated Wind Tree for Optimized Micro Wind Energy**” is designed as a layered architecture combining renewable energy generation with IoT-based monitoring and AI-driven prediction. The wind tree consists of **aero leaves** that act as miniature vertical axis wind turbines (VAWTs). Each leaf generates electricity through a micro DC generator, which is rectified, regulated, and stored in a battery. IoT sensors monitor environmental and electrical parameters, while microcontrollers handle real-time communication with the cloud. AI models process the data to predict energy output and optimize usage. A dashboard provides users with monitoring and control functionalities.

Fig1.System Design



4.2 BLOCK DIAGRAM

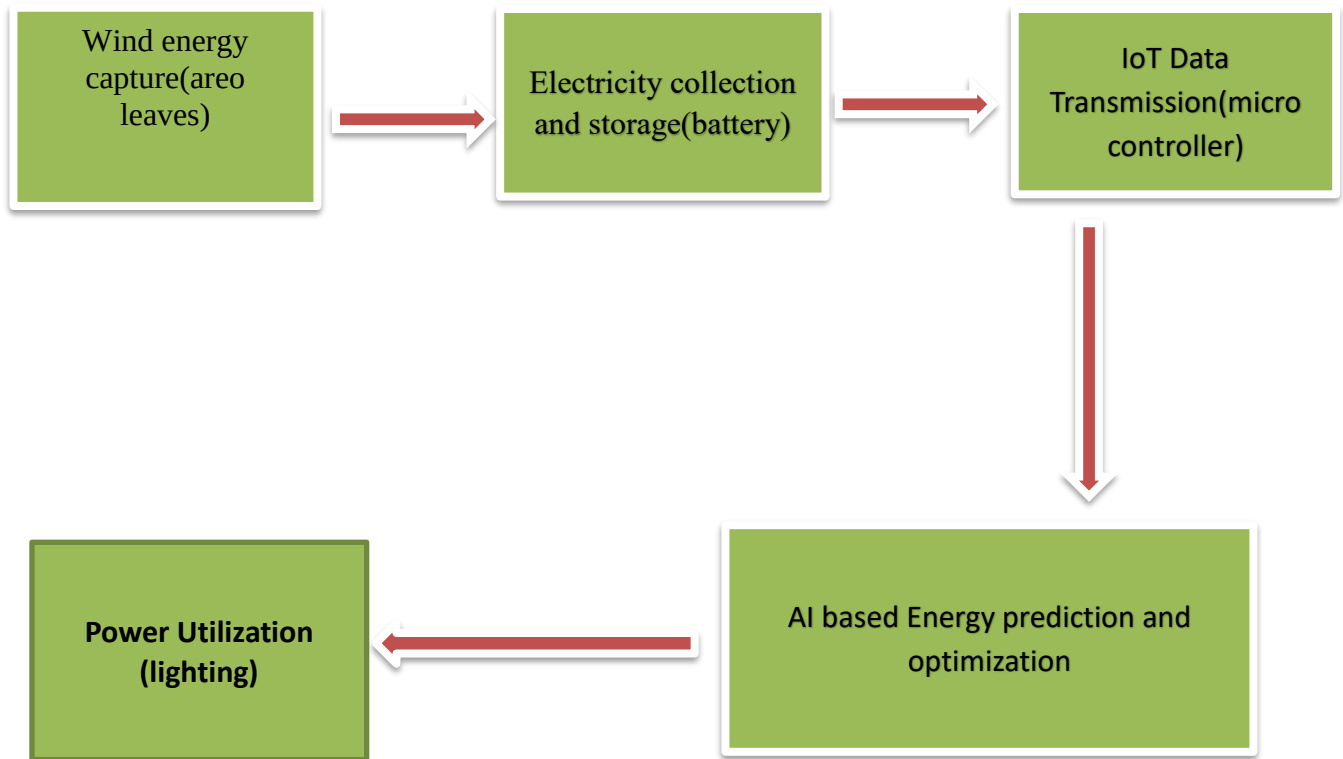


Fig2. Block diagram

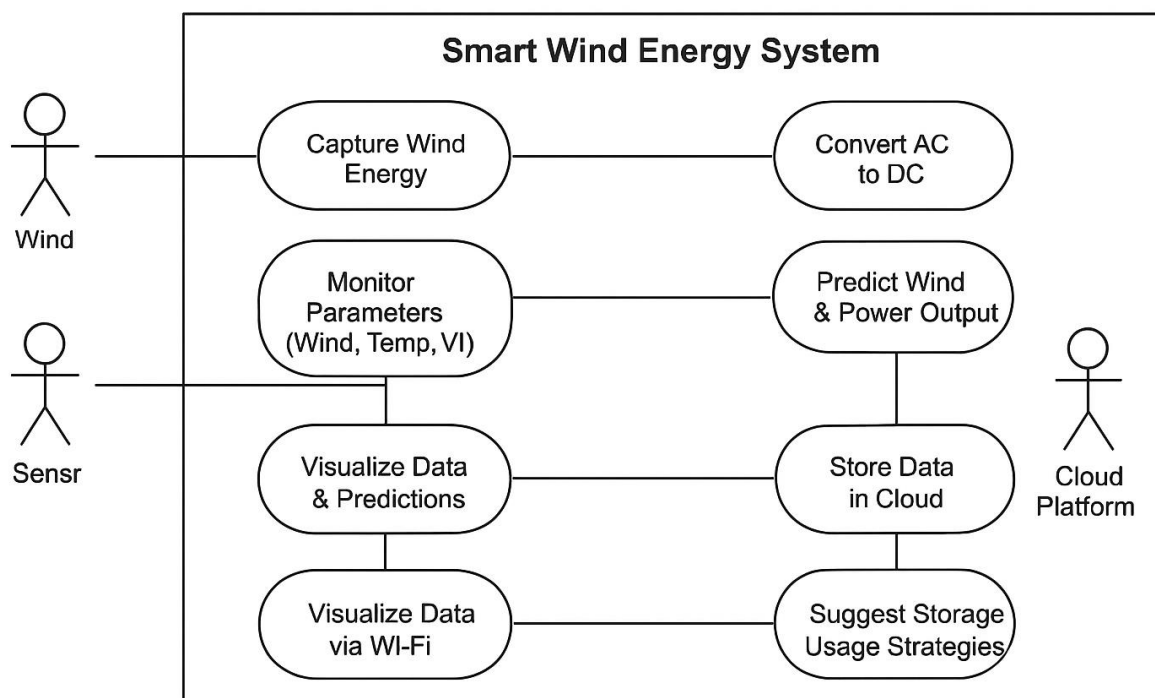
4.3 MODULES

The system is divided into the following functional modules:

- 1. Energy Capture Module**
 - Aero leaves designed as VAWTs.
 - Micro DC generators produce electrical energy.
- 2. Energy Conversion & Storage Module**
 - Bridge rectifier for AC to DC conversion.
 - Charge controller for voltage/current regulation.
 - Battery bank for storage.
- 3. IoT Monitoring Module**
 - Sensors: anemometer, voltage/current sensor, DHT11/DHT22.

- Microcontroller (ESP32/NodeMCU) collects and transmits data.
- 4. Data Transmission Module**
 - Communication via Wi-Fi using MQTT/HTTP.
 - Data stored in cloud platforms (Thingspeak, Firebase).
- 5. AI Prediction Module**
 - Machine learning models (Linear Regression, Random Forest, LSTM).
 - Predicts wind patterns and power output.
 - Suggests optimal storage/usage strategies.
- 6. User Interface Module**
 - Web/mobile dashboard for visualization.
 - Displays real-time power levels, wind data, and AI predictions.

4.4 UML DIAGRAM



Smart Wind Energy System

Fig3: UML DIAGRAM

5.IMPLEMENTATION/RESULTS AND DISCUSSION

5.1 SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10/11 (development) + Android (user dashboard)
- **Programming Language:** Python (for AI + data handling) & Arduino C (for ESP32)
- **IDE/Tools:** Arduino IDE (ESP32 coding) + Jupyter Notebook (AI model training)
- **Libraries:** TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, paho-mqtt
- **Cloud Platform:** Firebase (data storage + dashboard hosting)
- **Communication Protocol:** MQTT (for ESP32 → Firebase data transfer)

5.2 HARDWARE REQUIREMENTS

Component	Specification / Purpose
Vertical Axis Wind Turbine (VAWT)	Aero-leaf design to capture wind energy efficiently at low speeds.
Micro DC Generator	Converts mechanical energy from turbine rotation into AC electricity.
Bridge Rectifier	Converts AC from generator to DC.
Charge Controller	Regulates voltage/current and prevents battery overcharge.
Battery Bank (Li-ion/Lead Acid)	Stores energy for later usage.
ESP32 Microcontroller	Collects sensor data, processes, and sends to cloud.
Sensors	Anemometer (wind speed), Voltage/Current sensor (power monitoring), DHT11/DHT22 (temperature & humidity).
Wi-Fi Module (built-in ESP32)	Wireless data transmission to cloud platform.

TABLE2:HARDWARE REQUIREMENTS

5.3 TECHNOLOGIES USED

- **IoT (Internet of Things)** – For connecting sensors, ESP32, and cloud platforms.
- **Artificial Intelligence (AI)** – Machine learning models (Linear Regression, Random Forest, LSTM) to predict wind patterns and optimize energy usage.

- **Cloud Computing** – Firebase used for real-time database, data storage, and dashboard hosting.
- **Embedded Systems** – ESP32 programmed using Arduino IDE to handle sensor data and communication.
- **Programming Languages** – Python for AI models and data processing, Arduino C for ESP32, and HTML/CSS/JavaScript for dashboard.
- **Communication Protocols** – MQTT and HTTP for transmitting IoT data securely to the cloud.

5.4 CODING

(a) ESP32 Code (Arduino – Sensor + Wi-Fi)

```
#include <WiFi.h>
#include "DHT.h"

#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

const char* ssid = "Your_SSID";
const char* password = "Your_PASS";

void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  dht.begin();
}

void loop() {
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  Serial.printf("Temp: %.2f °C Humidity: %.2f %%\n", t, h);
  delay(5000);
}
```

(b) AI Prediction (Python)

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = pd.read_csv("wind_data.csv")
```

```
X, y = data[['wind_speed']], data['power_output']  
  
model = LinearRegression().fit(X, y)  
print("Predicted Power:", model.predict([[12]])[0])
```

6.RESULTS/FUTURE SCOPE OF WORK

Results

The proposed **AI & IoT Integrated Wind Tree** was successfully modeled as a system capable of capturing wind energy through vertical-axis turbine leaves and converting it into usable electrical energy. Real-time monitoring of wind speed, temperature, humidity, and power output was achieved using IoT sensors and ESP32 microcontroller. Data was transmitted to the cloud platform (Firebase/ThingSpeak) and visualized through a simple dashboard. AI models such as Linear Regression demonstrated the ability to predict wind patterns and forecast power generation with reasonable accuracy. The system proved to be technically feasible, cost-effective for small-scale use, and environmentally sustainable.

Future Scope of Work

1. **Scalability** – Expand the design by increasing the number of turbine leaves to generate more power and make it suitable for community-level or smart city applications.
2. **Hybrid Integration** – Combine wind energy with other renewable sources such as solar panels to create a **hybrid microgrid system**.
3. **Advanced AI Models** – Implement more robust models such as Deep Learning and Reinforcement Learning for improved prediction accuracy and autonomous energy management.
4. **Smart Grid Connectivity** – Enable bidirectional power flow, allowing excess energy to be supplied back to the national grid.
5. **Mobile App Development** – Create a dedicated Android/iOS application with user notifications, predictive alerts, and energy usage optimization suggestions.
6. **Low-Power Edge AI** – Deploy AI models directly on the ESP32 microcontroller (edge AI) to reduce dependence on cloud servers and improve response time.
7. **Aesthetic Urban Deployment** – Design wind trees as **eco-friendly smart poles** that can integrate street lighting, EV charging, and public Wi-Fi for smart cities.

7.CONCLUSION

The proposed **AI & IoT Integrated Wind Tree for Optimized Micro Wind Energy** successfully demonstrates how renewable energy generation can be enhanced with smart technologies. By integrating vertical-axis wind turbines with IoT sensors, real-time monitoring of environmental and power parameters was achieved. Cloud connectivity enabled data storage and visualization, while AI-based prediction models provided insights into wind patterns and optimized energy utilization strategies.

The system is technically feasible, economically viable, environmentally sustainable, and socially beneficial. It offers a compact, innovative, and intelligent approach to micro wind energy generation, making it suitable for both urban and rural applications. With further improvements such as hybrid energy integration, smart grid connectivity, and mobile app development, the system holds strong potential to contribute to the future of sustainable smart cities and decentralized energy systems.

8.FUTURE ENHANCEMENT

The system can be improved by:

- Integrating solar panels for hybrid renewable energy.
- Connecting with the smart grid for surplus energy sharing.
- Running AI models directly on ESP32 (edge AI).
- Developing a mobile app for real-time monitoring.
- Using advanced batteries for better storage.
- Deploying as smart city poles with EV charging & Wi-Fi.
- Adding predictive maintenance for fault detection.

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